

Database Management Systems

CO1 — Technical Presentation

ER Diagrams • EER Modeling • Data Independence • DBMS Architecture • Entity Types

NAME : K.HEMANTH
REG NO: (192572074)

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Library ER Diagram

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Data Independence

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DBMS Software Components

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Strong vs Weak Entities



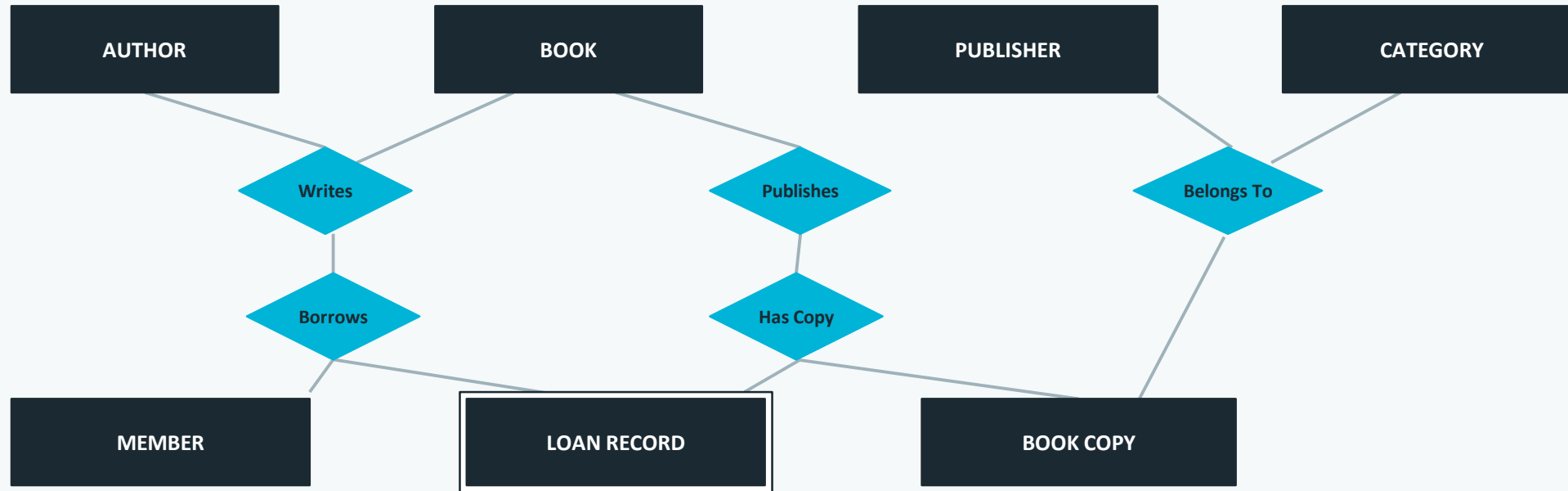
QUESTION 1

BL2 – UNDERSTAND

Create and present an ER diagram for an Online Library System, explaining each component clearly.

 Covered on the following slides

Online Library System — ER Diagram



Legend: ■ Entity ◆ Relationship ▭ Double Border = Weak Entity

Component Breakdown — Entities, Attributes & Relationships

MEMBER

MemberID (PK), Name, Email, JoinDate,
MembershipType

Represents each registered library user.

BOOK

ISBN (PK), Title, Edition, Language

The catalog record for a title, independent of physical copies.

BOOK COPY

CopyID (PK), BookISBN (FK), Condition, ShelfNo

An individual physical copy of a book available for lending.

AUTHOR

AuthorID (PK), Name, Nationality

Writer(s) linked to one or more books via Writes.

PUBLISHER

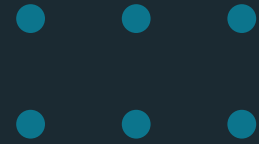
PublisherID (PK), Name, Address

Publishing house linked to books via Publishes.

LOAN RECORD

LoanID (partial key), MemberID (FK), CopyID (FK),
IssueDate, DueDate

Weak entity — cannot exist without a Member and a Book Copy.



QUESTION 2

BL2 – UNDERSTAND

Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.

Covered on the following slides

Generalization vs. Specialization — Quick Refresher



Generalization

Bottom-Up

Combines two or more entities that share common attributes into one higher-level super-class.

Example

UNDERGRADUATE + POSTGRADUATE → STUDENT



Specialization

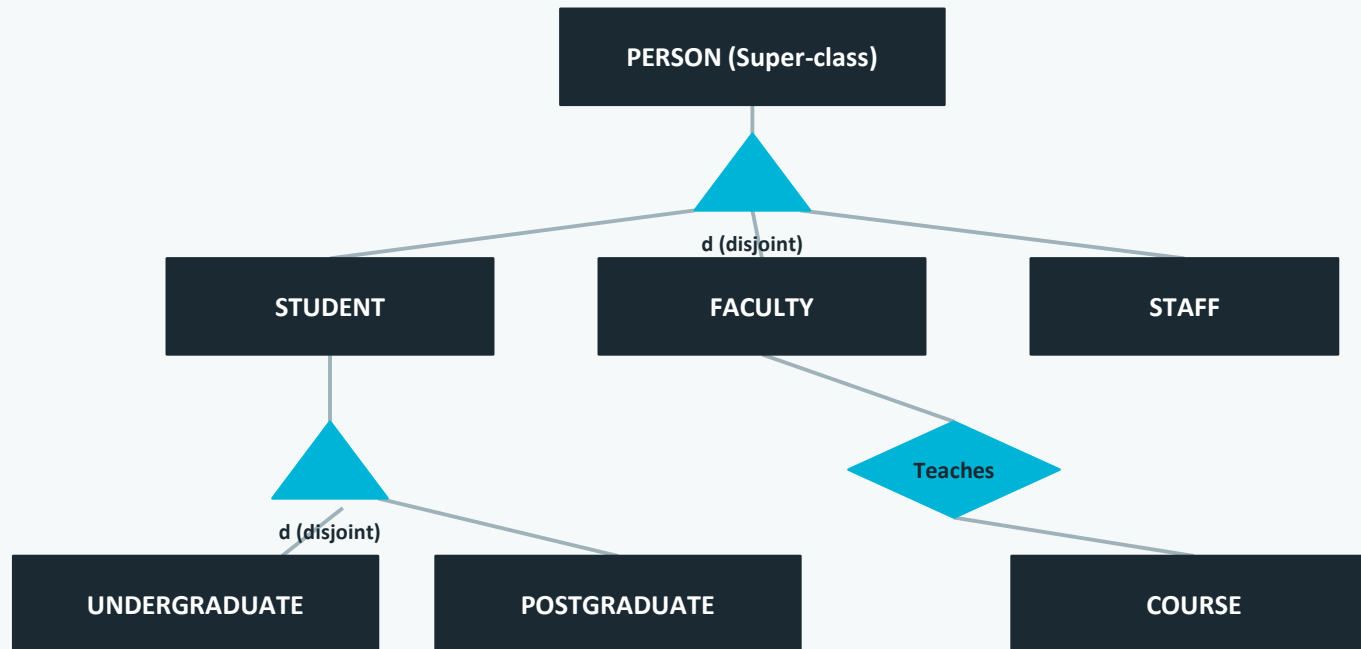
Top-Down

Divides a higher-level entity into distinct sub-classes based on unique characteristics.

Example

PERSON → STUDENT, FACULTY, STAFF

University Case Study — EER Diagram



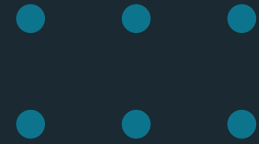
Legend: ▽ Triangle = ISA / Specialization

Two Levels of Specialization

PERSON specializes into STUDENT, FACULTY, and STAFF — each inherits Name, DOB, and Address, adding role-specific attributes (e.g. Faculty adds Department).

STUDENT is further specialized into UNDERGRADUATE and POSTGRADUATE — a second-level (multi-level) specialization based on programme type.

Constraint used: Disjoint, Total — every Person is exactly one of Student, Faculty, or Staff.



QUESTION 3

BL2 – UNDERSTAND

Present the concept of data independence (logical and physical) and its significance in DBMS architecture.

Covered on the following slides

The Three-Schema Architecture & Data Independence

EXTERNAL LEVEL

User views — customized presentation for each user group

Logical Data Independence

Logical: change the conceptual schema (e.g. add a new attribute) without altering external views or application programs.

CONCEPTUAL LEVEL

Logical schema — entities, relationships, and constraints

Physical Data Independence

Physical: change storage structures or file organization without altering the conceptual schema or application programs.

INTERNAL LEVEL

Physical schema — storage structures, indexes, and file organization

Why Data Independence Matters

1

Reduced Maintenance

Schema changes don't ripple into every application, cutting redevelopment cost and time.

2

Easier Evolution

New requirements — new attributes, tables, or storage engines — can be added with minimal disruption.

3

Improved Security

External views expose only relevant data, hiding the full logical and physical structure from users.

4

Storage Flexibility

Indexes, file formats, or hardware can be optimized or migrated without touching application code.

5

Vendor / Platform Freedom

Physical independence eases migration between storage engines or hardware platforms.

6

Better Data Integrity

Centralizing structure in the conceptual schema keeps constraints consistent across all user views.



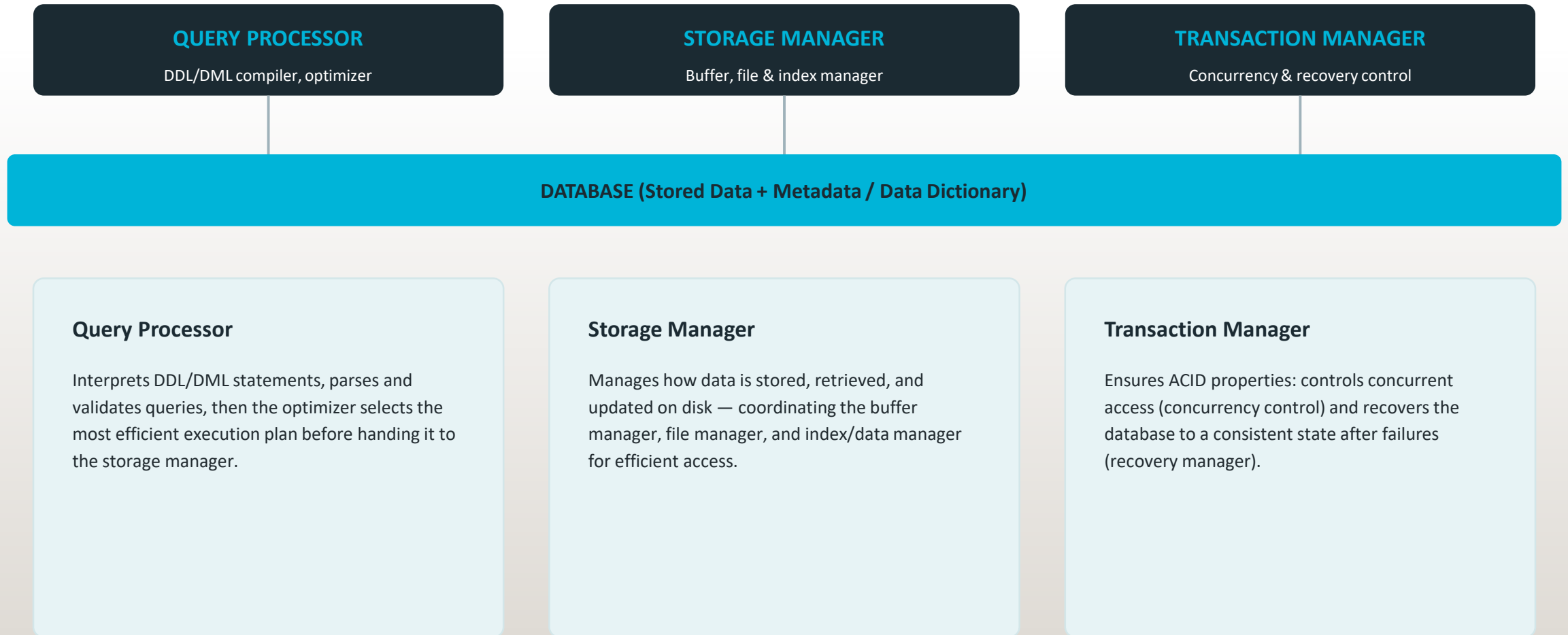
QUESTION 4

BL1 – REMEMBER

Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.

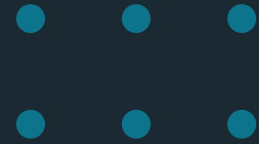
Covered on the following slides

DBMS Software Components — High-Level Architecture



Component Sub-Modules at a Glance

Module	Sub-Component	Function
Query Processor	DDL Interpreter	Processes schema definitions and stores them in the data dictionary.
Query Processor	DML Compiler	Translates DML statements into low-level instructions the query evaluator understands.
Query Processor	Query Optimizer	Chooses the lowest-cost execution plan among equivalent relational algebra expressions.
Storage Manager	Buffer Manager	Controls transfer of data between main memory and disk, deciding what to cache.
Storage Manager	File Manager	Manages allocation of disk space and the data structures for representing stored data.
Transaction Manager	Concurrency Control	Ensures simultaneous transactions execute without conflicting or corrupting data.
Transaction Manager	Recovery Manager	Restores the database to a consistent state after a system crash or error.



QUESTION 5

BL2 – UNDERSTAND

Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.

Covered on the following slides

Strong Entities vs. Weak Entities



Strong Entity

- Has its own primary key attribute
- Exists independently of any other entity
- Represented with a single-line rectangle

Examples: STUDENT (StudentID), ORDER (OrderID), EMPLOYEE (EmpID)

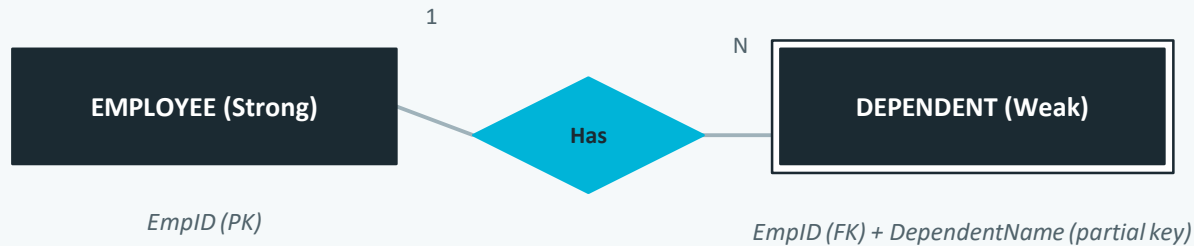


Weak Entity

- Has no standalone primary key — only a partial key
- Existence depends on a related (owner) strong entity
- Represented with a double-line rectangle

Examples: DEPENDENT (owned by Employee), ORDER_ITEM (owned by Order)

Real-World Example — Employee & Dependent



A DEPENDENT (e.g. an employee's child, for insurance purposes) cannot be identified or exist in the database without knowing which EMPLOYEE they belong to. Its identifying key is the combination of the owner's key (EmpID) and its own partial key (DependentName).

Quick Comparison

Identification

Strong: Own primary key | **Weak:** Owner key + partial key

Existence

Strong: Independent | **Weak:** Dependent on owner

ER Notation

Strong: Single rectangle / diamond | **Weak:** Double rectangle / diamond

Summary

- 1 The Online Library System ER diagram links Member, Book, Author, Publisher, Category, and the weak Loan Record entity.
- 2 The EER model applies two-level specialization in a University case study: Person → Student/Faculty/Staff, then Student → Undergraduate/Postgraduate.
- 3 Logical and physical data independence let schema and storage change without breaking applications — reducing maintenance and easing evolution.